

pulses 

$$\mathbf{x}_i = (\text{charge}_i, \text{dom_time}_i, \dots)$$



each pulse $\rightarrow$ a vector $\mathbf{z}_i^{(0)}$



transformer block (repeated $\times L$)

multi-head self-attention

h heads
in parallel

$$\mathbf{q}_i = W_Q \mathbf{z}_i$$

$$\mathbf{k}_i = W_K \mathbf{z}_i$$

$$\mathbf{v}_i = W_V \mathbf{z}_i$$

query, key, value
(learned W_Q, W_K, W_V)

$$a_{ij} = \frac{\exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}{\sum_{\ell} \exp(\mathbf{q}_i \cdot \mathbf{k}_{\ell} / \sqrt{d_k})}$$

how strongly
pulse i attends
to pulse j

$$\tilde{\mathbf{z}}_i = \sum_j a_{ij} \mathbf{v}_j$$

new pulse vector,
a weighted sum
of values

residual



add residual & normalise

keep values
well behaved



feed-forward network (MLP)

transform each
pulse on its own



add residual & normalise

residual



readout: pool pulses or read a CLS token

event $\rightarrow$ one vector



task-dependent prediction head

$$s(\mathcal{E}) = P(y=1 \mid \mathcal{E})$$

classification

$$\hat{\mathbf{n}} \in S^2$$

direction regression

$$(\hat{\mathbf{n}}, \kappa)$$

vMF (direction + uncertainty)